

Compact Representations of Human Interception Movements

Low-dimensional models for representing and generating human interception movement

Seman Libbiss & Paz Flashner

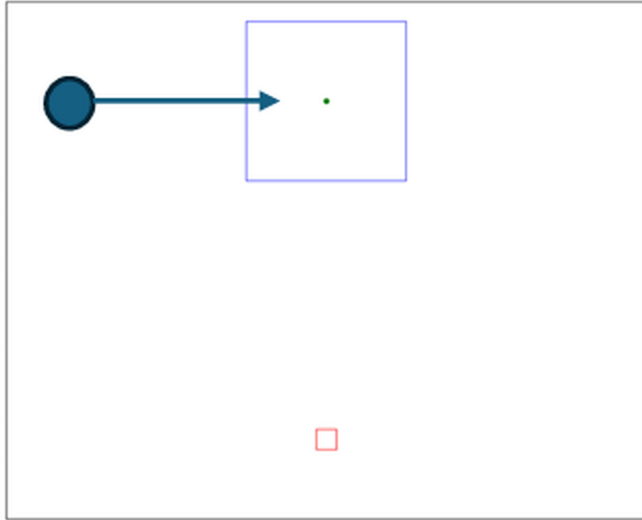
Research supervision: Prof. Jason Friedman | Course advisor: Moni Shahrar

Workshop on Deep Learning, Tel Aviv University

Planned talk: 10 minutes (11 core slides + backup slides)

The experimental task and the data structure

50 s | 2/11



The experimental task

- **Participant's task:** move the finger from the lower start zone into the upper interception square exactly as the ball passes through it.
- **Conditions:** the free eye-movement condition is analysed. Six target start locations (3 right, 3 left) over a continuous speed range.
- **Sample:** 4,732 valid trials from 28 participants, measured in 3D at 240 Hz with a Polhemus system.

A strict split that prevents information leakage

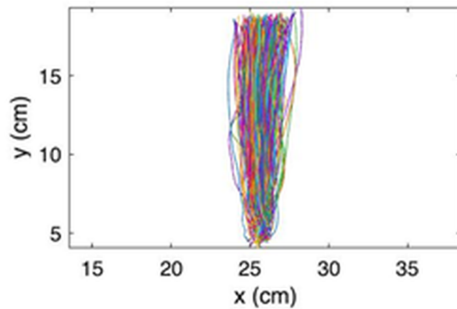
Four fixed folds: 17 training participants, 4 validation and 7 test. Every participant appears in a test fold exactly once.

What we set out to answer

40 s | 3/11

The core question

Human movement is stochastic: perception and motor control vary from trial to trial. One participant, one target - and every trial takes a different path.



Can a compact fingerprint capture that spread?

Compress 200 numbers - the 100 x 2 path - into a 3-to-8 dimensional fingerprint of an individual's movement distribution.

100 x 2 coordinates $\rightarrow$ $n = 3$ or 8 latent numbers (durations predicted)

What would count as success, evaluated on participants the model never trained on:

SUCCESS TEST 1

Reconstruct

Rebuild specific recorded trajectories from the compressed code.

SUCCESS TEST 2

Generate

Sample new trajectories matching that person's stochastic distribution.

Trajectory processing and the compression architectures

65 s | 4/11

Stage in the model	Spline + PCA (linear baseline)	Neural models (VAE / CVAE / CAE)
Primary input	The same 100 x 2 normalised trajectory	The same 100 x 2 trajectory + 5 task conditions
Where standardisation happens	On the 18 spline coefficients, after fitting	On the 200 coordinates, with train-set constants
Encoder / compression	Cubic B-spline fit, 9 coefficients per axis -> PCA to n latents	Two fully connected 256-unit ReLU layers -> latent n
Decoder / reconstruction	Inverse PCA -> multiply by the spline basis for the full path	Two 256-unit ReLU layers -> 200 coordinates + 2 timing branches
Timing prediction	Separate Ridge regression on the PCA scores + conditions	Dedicated head predicting initiation and movement time

A fair comparison at identical dimensions

Models are compared at identical latent dimensions of $n = 3$ (extreme compression) and $n = 8$ (extended capacity).

Task 1: reconstructing a recorded trajectory (Reconstruction)

Key insight on pointwise reconstruction

PCA wins at reconstructing a single trajectory.

The linear PCA projection is optimal by definition for minimising squared error. VAE models pay in pointwise accuracy to obtain a continuous, regular and samplable latent space.

Trajectory reconstruction error (MSE)

The model receives a fully observed trajectory, compresses it to n values and tries to rebuild it:

Model	MSE $n=3$	MSE $n=8$
Spline + PCA	0.0972	0.0228
VAE (Unconditional)	0.1483	0.0335
Conditional AE (CAE)	0.1697	0.0260
CVAE	0.1770	0.0325
Condition-only Ridge (Floor)	1.2338	1.2338

Task 2: generating a personal movement distribution

60 s | 6/11

11 kinematic features

Initiation and movement time, peak speed, time to peak, path length, curvature, maximum lateral deviation and endpoints.

Distribution comparison metrics

KS tests for the marginals, alongside energy discrepancy and MMD squared for the joint structure of all 11 features.

How do we generate movements from a fingerprint?

- **Enrollment:** earlier context trials of the participant, whose mean defines the personal fingerprint centre.
- **Stochastic sampling:** draw 120 latent vectors around that centre using a shared covariance learned from training data.
- **Unseen movements:** query trials are held aside to evaluate the distribution and are never exposed to the model.

$$\theta_s = (1 / |C_s|) \sum \mu_i z(b) \sim N(\theta_s, \Sigma_{\text{train}})$$

Generation results: VAE leads on movement distributions

Model	n	Mean KS (lower = better)	Energy discrepancy	MMD squared
Spline + PCA	3	0.3251	0.9774	0.1815
CVAE	3	0.2778	0.7454	0.1309
VAE (Unconditional)	3	0.2846	0.7476	0.1303
Spline + PCA	8	0.2548	0.7347	0.1153
CVAE	8	0.2194	0.4305	0.0675
VAE (Unconditional) - default	8	0.2151	0.4110	0.0639

A reversal in the ranking: VAE wins at generation

Although Spline + PCA won pointwise reconstruction, VAE and CVAE achieve much better distribution matching (corrected $p < 0.001$ under BH and Holm). VAE at $n = 8$ was set as the dashboard default.

Validating the personal fingerprint (Control Test)

27 / 28

participants improved markedly

Using the personal fingerprint improved distribution matching against generic attribution ($p < 1.5 \times 10^{-6}$).

Does the personal fingerprint really matter?

To verify that the model is not merely producing a generic average movement, we replaced the fingerprint while the decoder, the noise and the conditions stayed completely frozen:

Injected fingerprint centre	CVAE n=8 (Mean KS)	VAE n=8 (Mean KS)
The participant's own centroid	0.2194	0.2151
Training-population mean	0.2896	0.2841
Another test participant	0.3172	0.3144

Recommended summary: the trade-off between reconstruction

45 s | 9/11

To reconstruct known trajectories

Spline + PCA

Recommended when the goal is compact compression and the most accurate pointwise reconstruction of a measured trajectory (MSE = 0.0228).

For generation and simulation

VAE (n = 8)

Recommended for simulating a participant's future movements. Produces the kinematic distributions closest to reality (MMD squared = 0.0639).

! Key caveat

Direct measurement from the participant's own context data still beats latent regressions at predicting pointwise summary measures. The compact representation is meant to generate whole trajectories, not a single number.

The applied product: an interactive dashboard

80 s | 10/11

1

Model and dimension

Supports CAE, CVAE, VAE and Spline + PCA at dimensions 2, 3, 4 and 8.

2

Fingerprint selection

Slider control over the latent coordinates and real-time simulation of the resulting trajectory.

3

Distribution sampling

Generate 25 to 500 trajectories and compare them against the participant's real data.

Streamlit App

Run locally in a terminal:

```
python -m streamlit run src/confirmatory_dashboard.py --server.port 8510
```

Conclusions and open questions for the lab

30 s | 11/11

Open questions for discussion (Prof. Jason Friedman)

- **Stability across sessions:** does the fingerprint stay stable across different days or sessions?
- **Model window start:** should the window begin at target appearance, or at the start of target motion?
- **Consistent timing scaling:** should physical times be explicitly bounded inside the model?

Summary of findings

A compact eight-value representation preserves clear personal information and reproduces individual movement distributions.

There is a clear split between the model that is best for reconstruction (Spline + PCA) and the one that is best for generation (VAE).

Backup: resampling and the two time windows

BACKUP

Model input: target motion to final tracker sample.

Feature input: detected finger onset to end.

A phase of 0.5 means half the selected time interval, not half the path length.

$$\mathbf{P}_j = \mathbf{S}(j/99) - \mathbf{S}(0), \quad j = 0, \dots, 99.$$

100 samples are a common representation, not 100 independent observations.

Backup: what becomes the PCA matrix?

[BACKUP](#)

Object	Dimensions	One row / one column means
Sampled trial	100 x 2	Time point / x or y
Spline basis B	100 x 9	Time point / basis function
Trial coefficient vector	18	Nine x coefficients followed by nine y
Training coefficient matrix	N training x 18	Trial / coefficient
PCA directions V	18 x n	Coefficient / retained component
Training latent codes	N training x n	Trial / latent coordinate

Five interior knots + degree 3 + 1 = nine coefficients per axis.

Backup: fitting and decoding spline + PCA

BACKUP

Fit the two coordinate coefficient vectors by least squares.

Validation selects raw or standardized coefficients.

Inverse PCA gives coefficients; multiplying by B restores the path.

$$\mathbf{z} = V^T[(\mathbf{c} - \mathbf{m}) \oslash \mathbf{d}], \quad \hat{\mathbf{c}} = \mathbf{m} + \mathbf{d} \odot (V\mathbf{z}).$$

m and d combine centering/scaling; V contains retained PCA directions. Raw PCA uses d = 1.

Backup: predicting physical timing

BACKUP

Stage	Spline + PCA	Neural models
Predictors	n PCA scores + 5 conditions	Latent code + conditions where enabled
Targets	Two standardized $\log(t + .001)$ values	Same two standardized targets
Predictor	Ridge, alpha selected on validation	Separate decoder output head
Inverse	Undo scaling, exponentiate, subtract .001	Same inverse transform

Observed timing never enters the trajectory encoder. Reconstruction-time timing prediction sees the observed shape.

Backup: the neural training objective

BACKUP

Coordinate error sums 200 squared differences.

Timing error sums two squared differences and receives weight 20.

KL sums across latent dimensions; the batch averages the total.

$$\mathcal{L} = \|\mathbf{v} - \hat{\mathbf{v}}\|_2^2 + 20\|\mathbf{y} - \hat{\mathbf{y}}\|_2^2 + \beta D_{\text{KL}}[q(\mathbf{z} \mid \mathbf{v}, \mathbf{c}) \parallel \mathcal{N}(\mathbf{0}, I)].$$

Replacing both summed terms by separate MSE means would change the effective timing weight.

Backup: VAE, CVAE and CAE

BACKUP

Property	VAE	CVAE	CAE
Trajectory encoder	Two hidden layers	Two hidden layers	Two hidden layers
Task inputs	Masked to zero	Encoder and decoder	Encoder and decoder
Latent training code	Gaussian sample	Gaussian sample	Deterministic
KL penalty	Yes	Yes	No
Reconstruction code	Posterior mean	Posterior mean	Deterministic code

A nonsignificant conditional-model advantage does not mean the experimental conditions are irrelevant.

Backup: where the generated variation comes from

BACKUP

The personal center averages context codes.

Training residuals estimate within-person latent variation.

Variational models also add average posterior variance.

$$\Sigma_{\text{train}} = \text{Cov}_{\text{train}}(\boldsymbol{\mu}_i - \bar{\boldsymbol{\mu}}_{s(i)}) + \text{diag}(\mathbb{E}_{\text{train}}[\boldsymbol{\sigma}_i^2]) + 10^{-6}I.$$

One shared covariance for every participant; 120 draws around each personal center.

Backup: leakage and held-out evaluation

BACKUP

Information	Training	Validation	Test context	Test query
Fit network/PCA/scalers	Yes	No	No	No
Select checkpoint/settings	No	Yes	No	No
Personal center	No	No	Yes	No
Reconstruction input	No	No	No	Observed path
Generation task mixture	No	No	No	Metadata only
Generation scoring	No	No	No	Paths and timings

Population model fitting and participant enrollment are separate stages.

Backup: what the eleven features measure

BACKUP

Group	Features
Timing	Initiation time; movement time; relative time to peak speed
Speed	Peak speed; heuristic speed-peak count
Geometry	Path length; straight-line distance; curvature ratio; maximum lateral deviation
Endpoint	Final x; final y

Generated feature computation uses predicted timing and an onset-based movement crop.

Backup: KS and joint distribution distances

BACKUP

KS: largest gap between two empirical cumulative curves.

Mean KS: average of eleven feature statistics.

Energy and MMD: compare whole vectors and account for joint structure.

$$D_{sf} = \sup_x |F_{sf}(x) - G_{sf}(x)|,$$

$$\overline{D}_s = \frac{1}{11} \sum_{f=1}^{11} D_{sf}.$$

KS statistics are not p-values. Correlated features can affect joint geometry.

Backup: energy discrepancy and mean collapse

[BACKUP](#)

Energy compares cross-set distances with within-set distances.

A generator needs plausible variation as well as a good center.

Example reference	Mean-path RMSE	Path energy
VAE n=8	0.1604	0.0548
Repeated context mean	0.1146	0.4621

$$E = 2 \overline{\|X - Y\|} - \overline{\|X - X'\|} - \overline{\|Y - Y'\|}.$$

Lower is better. The mean-only output has a closer mean but a much worse distribution.

Backup: Gaussian-kernel MMD

BACKUP

Kernel: $\exp(-\gamma \times \text{squared Euclidean distance})$.

Gamma comes from the median positive training pair distance.

Within-set diagonals are excluded in the unbiased estimator.

$$\widehat{\text{MMD}}_u^2 = \frac{\sum_{i \neq j} k(X_i, X_j)}{m(m-1)} + \frac{\sum_{i \neq j} k(Y_i, Y_j)}{r(r-1)} - \frac{2 \sum_{i,j} k(X_i, Y_j)}{mr}.$$

Finite-sample unbiased MMD squared may be slightly negative. It is not an accuracy percentage.

Backup: 28 paired scores, not thousands of independent trials

BACKUP

First average neural seeds within each participant.
Then compare paired participant scores using two-sided Pratt Wilcoxon.
Round differences to 12 decimals before ranking.

Correction family	Tests
Original seven endpoints x ten pairs x two dimensions	140
New two path endpoints x ten pairs x two dimensions x two geometries	80
VAE replacement controls	6
CVAE replacement controls	12

BH controls false discoveries under its assumptions. Holm is a familywise-error sensitivity. Nonsignificance is not equivalence.

Backup: the boundaries of the conclusion

BACKUP

Useful same-session personal centers do not establish stable traits.

Direct context summaries beat latent linear probes on all 14 tested targets.

Condition routes work, but their value depends on the task and model.

Minimum-jerk decomposition remains a separate investigation.

The 31-page laboratory report retains the extended results and Jason's open questions.

Backup: launch and source navigation

BACKUP

Repository terminal:

```
python -m streamlit run src/confirmatory_dashboard.py --server.port 8510
```

Browser: <http://127.0.0.1:8510>

Stop: Ctrl+C in that terminal

The standalone dashboard archive contains launch instructions for use without the full research checkout.